

Curriculum Vitae

Current position

01.2024 – present **Team leader**
 Paris Brain Institute
 Neuronal circuits and brain dynamics team
 Tenured researcher (Chargé de recherche), INSERM
 ERC StG 2024 Laureate

Nikolaos Karalis

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Professional experience

11.2022 – present **Independent team leader**
 Friedrich Miescher Institute for Biomedical Research
Ambizione fellow

01.2018 – 10.2022 **Research fellow**
 Friedrich Miescher Institute for Biomedical Research
 Project: *Neuromodulatory organization of amygdala circuits*
 EMBO LTF and MSCA IF fellow
 Group leader: Prof. Dr. Andreas Lüthi

Education

09.2013 - 12.2017 **Ph.D. in Neuroscience**
 Faculty of Medicine, Ludwig-Maximilians University Munich
 Thesis: *Oscillatory architecture of memory circuits* (Thesis defense: 25.02.2019)
 Advisor: Prof. Dr. Anton Sirota

2011 - 2013 **M.Sc. Neurasmus Joint Master in Neuroscience**
 09.2011 - 09.2012 Charité University Medicine, Berlin – Medical Neuroscience
 09.2012 - 08.2013 University Bordeaux II – Neuroscience and Neuropsychopharmacology
 Thesis: *Neuronal signatures of fear memory*

09.2004 – 03.2011 **B.Sc. & M.Sc.**
 School of Applied Mathematics and Physics, National Technical University of Athens (NTUA)
 Majors: Computational Mathematics, Statistics
 Thesis: *EEG signal analysis methods for the characterization of meditative states*

Scholarships and Grants

01.2025 – ERC Starting Grant (1.5M €)
 01.2025 – Foundation Schlumberger for Education and Research award 2024 (100k€)
 01.2024 – Tenured Inserm Chargé de recherche (CR)
 01.2024 – Paris Brain Institute endowment
 01.2024 – 01.2028 ATIP-Avenir (500k €)
 01.2024 – 01.2026 BBRF NARSAD (\$70k)
 11.2022 – 01.2025 Swiss National Foundation Ambizione (835k CHF)
 06.2021 – 10.2022 Marie Curie Individual Fellowship (191k €)
 04.2019 – 06.2021 EMBO Long-Term Fellowship (151k CHF)
 12.2014 – 12.2017 Ludwig-Maximilians-Universität München
 09.2013 – 11.2014 Centre for Integrative Neuroscience (CIN) - University of Tübingen
 09.2011 – 08.2013 Erasmus Mundus scholarship for the *Neurasmus Joint Master degree program in Neuroscience*
 06.2010 University of Pennsylvania Grant
 03.2009 Erasmus scholarship (Research exchange)
 07.2007 MITACS scholarship

Awards

2024 Foundation Schlumberger for Education and Research award
2018 Onassis Foundation nomination and grant to attend Lindau Nobel Laureate Meeting
2010 “*Thomaideio Award*” for best conference presentation from NTUA
2008 “*Thomaideio Award*” for best journal publication from NTUA

Conference organization

2026	FENS Forum 2026 symposium “ <i>Breathing as a dynamic scaffold of learning, emotion, and neural dynamics</i> ”
2026	Organizer of Paris Circuit Dynamics 2 “ <i>Neuromodulators</i> ” conference
2025	Organizer of Paris Circuit Dynamics 1 “ <i>Hippocampal–Cortical Circuits</i> ” conference
2020 - 2023	Organizer of FMI Young Researcher seminar series (>80 speakers hosted)
2017	Co-organizer of Miniscope Technology Transfer Workshop at LMU, Munich
2015 - 2017	Organizer of Neurophysiology Nights seminar series, LMU
2012	Organizer of “Speaking to the public” (M.Sc. course), Charité University Hospital, Berlin

Teaching

2026	Diplôme Inter-Universitaire NeuroTrans 2025 (Paris-Saclay, Sorbonne)
2025 - present	Thesis committees
2025 - present	CNR ERC Mentor Programme
2024 - present	Student tutoring and hosting in the context of Stanford exchange programs
2024	Student tutoring for course “Hot Topics in Neurodegenerative and psychiatric diseases” (Sorbonne Uni)
2019, 2024	Tutor at FENS CAJAL course - Biosensors and actuators for systems neuroscience, Bordeaux
2015	Tutor at the 7th G-Node Winter Course in Neural Data Analysis

Academic service & outreach

2025 – present	Scientific coordinator of France Brain Bee Olympiad
2024 – present	Scientific expert committee member (ICM Data Analysis Core, ICM RnD unit, ICM ephys platform)
2024 – present	Numerous undergraduate and high-school students hosted and tutored
2020 – 2023	FMI young investigator seminar series organizer (>80 speakers hosted)
2019 – 2023	FMI postdoc representative
2016 – present	Administrator of Systems Neuroscience mailing list (>2500 members)
2016 – present	Peer reviewer
2012 – 2015	Writer & editorial board of the “CNS Charité Neuroscience Newsletter”
2012 – 2013	Neurasmus course representative at the Erasmus Mundus Association (EMA)
2004 – 2006	Coordinator and invigilator - National mathematical competitions - Hellenic Mathematical Society (HMS)
2004	Coordinator, guide, and invigilator at the International Mathematical Olympiad (IMO) 2004
2004	Coordinator, guide, and invigilator at the International Informatics Olympiad (IOI) 2004

Invited Talks

2025	ICONIP2025 GDR Neural Net Paris Biological Physics Community Day Neural mechanisms of cognitive function French Neuroscience Society Swiss Society of Sleep Research Polish Center for Technology Development	2017	Janelia Junior Scientist Workshop on Neural Circuits Coupling & Causality in Complex Systems Bernstein Conference PhD Symposium Harvard-LMU Young Scientists' Forum Japan Neuroscience Society meeting British Neuroscience Association meeting
2024	Paris Neural Theory Symposium IFM Young Researchers' Day 2024	2016	LMU lecture series Neurizons – Young Investigator Talks
2023	Giessbach meeting	2015	German Neuroscience Society
2022	Paris Brain Institute FMI Annual Meeting	2014	Animal Communication Workshop
2021	Giessbach meeting	2013	Neurasmus Workshop Neurasmus Orientation Week
2020	Neurizons – Young Investigator Talks Neuromatch 2.0	2011	Breaking Convention
2019	FENS-Hertie Winter School	2007	Mitacs Industrial Math Summer School Canadian Undergr. Math. Conference

Full publication list**Publications in international peer-reviewed scientific journals**

1. Grimm C, ..., **Karalis N**, ..., Zerbi V
Tonic and burst-like locus coeruleus stimulation distinctly shift network activity across the cortical hierarchy
Nature Neuroscience, 2024
2. Massi L., Hagihara K., ..., **Karalis N.***, Lüthi A.*
Di-synaptic specificity of serial information flow for conditioned fear
Science Advances, 2023
3. **Karalis, N.**, & Sirota, A
Breathing coordinates cortico-hippocampal dynamics in mice during offline states.
Nature communications, 2022
4. Hagihara, K. M., Bukalo, O., Zeller, M., Aksoy-Aksel, A., **Karalis, N.**, Limoges, A., Rigg, T., Campbell, T., Mendez, A., Weinholtz, C., Mahn, M., Zweifel, L. S., Palmiter, R. D., Ehrlich, I., Lüthi, A., & Holmes, A.
Intercalated amygdala clusters orchestrate a switch in fear state.
Nature, 2021
5. Mahn, M., Saraf-Sinik, I., Patil, P., Pulin, M., Bitton, E., **Karalis, N.**, Bruentgens, F., Palgi, S., Gat, A., Dine, J., Wietek, J., Davidi, I., Levy, R., Litvin, A., Zhou, F., Sauter, K., Soba, P., Schmitz, D., Lüthi, A., Rost, B. R., ... Yizhar, O.
Efficient optogenetic silencing of neurotransmitter release with a mosquito rhodopsin.
Neuron, 2021
6. Castillo, X., Castro-Obregón, S., Gutiérrez-Becker, B., Gutiérrez-Ospina, G., **Karalis, N.**, Khalil, A. A., Lopez-Noguerola, J. S., Rodríguez, L. L., Martínez-Martínez, E., Perez-Cruz, C., Pérez-Velázquez, J., Piña, A. L., Rubio, K., García, H., Syeda, T., Vanoye-Carlo, A., Villringer, A., Winek, K., & Zille, M.
Re-thinking the Etiological Framework of Neurodegeneration.
Frontiers in neuroscience, 2019
7. Babaev, O., Cruces-Solis, H., Piletti Chatain, C., Hammer, M., Wenger, S., Ali, H., **Karalis, N.**, de Hoz, L., Schlüter, O. M., Yanagawa, Y., Ehrenreich, H., Taschenberger, H., Brose, N., & Krueger-Burg, D.
IgSF9b regulates anxiety behaviors through effects on centromedial amygdala inhibitory synapses.
Nature communications, 2018
8. Rozeske, R. R., Jercog, D., **Karalis, N.**, Chaudun, F., Khoder, S., Girard, D., Winke, N., & Herry, C.
Prefrontal-Periaqueductal Gray-Projecting Neurons Mediate Context Fear Discrimination.
Neuron, 2018
9. Dejean, C. *, Courtin, J. *, **Karalis, N. ***, Chaudun, F., Wurtz, H., Bienvenu, T. C., & Herry, C.
Prefrontal neuronal assemblies temporally control fear behaviour.
Nature, 2016
10. **Karalis, N.**, Dejean, C. , Chaudun, F. , Khoder, S., Rozeske, R. R., Wurtz, H., Bagur, S., Benchenane, K., Sirota, A., Courtin, J., & Herry, C.
4-Hz oscillations synchronize prefrontal-amygdala circuits during fear behavior.
Nature Neuroscience, 2016
11. Courtin, J., Chaudun, F., Rozeske, R. R., **Karalis, N.**, Gonzalez-Campo, C., Wurtz, H., Abdi, A., Baufreton, J., Bienvenu, T. C., & Herry, C.
Prefrontal parvalbumin interneurons shape neuronal activity to drive fear expression.
Nature, 2014
12. Courtin, J., **Karalis, N.**, Gonzalez-Campo, C., Wurtz, H., & Herry, C.
Persistence of amygdala gamma oscillations during extinction learning predicts spontaneous fear recovery.
Neurobiology of learning and memory, 2014
13. Falagas, M. E., Peppas, G., Matthaiou, D. K., Karageorgopoulos, D. E., **Karalis, N.**, & Theocharis, G.
Effect of meteorological variables on the incidence of lower urinary tract infections.
European journal of clinical microbiology & infectious diseases, 2009

Patents

WO/2017/021542

Method and device for modulating fear and/or anxiety

*: equal contribution